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United States Patent Application Publication

20250285078

Kind Code

A1

Publication Date

September 11, 2025

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System and Method of Partition Overload Handling for E-Commerce Available to Promise Inquiries

Abstract

A system and method are disclosed for handling an available to promise inquiry. The method comprises requesting statistics for partitions of a distributed cache, determining whether a threshold for the partitions has been reached, in response to a determination that the threshold for the partitions has been reached, fetching inventory data for the available to promise inquiry from a local cache, and in response to a determination that the threshold for the partitions has not been reached, fetching inventory data for the available to promise inquiry from the distributed cache. The method further comprises setting an expiry time before the local cache is purged, and storing a status of the partitions based at least in part on the determining that the threshold for the partitions has been reached, the status indicating the partitions as being overloaded.

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Family ID: 96056036

Appl. No.: 19/214682

Filed: May 21, 2025

Related U.S. Application Data

parent US continuation 18207930 20230609 parent-grant-document US 12333490 child US 19214682

us-provisional-application US 63423911 20221109

Publication Classification

Int. Cl.: G06Q10/087 (20230101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/207,930, filed Jun. 9, 2023, entitled “System and Method of Partition Overload Handling for E-Commerce Available to Promise Inquiries,” which claims the benefit under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/423,911, filed Nov. 9, 2022, entitled “Partition Overload Handling for E-Commerce ATP Inquiries.” U.S. patent application Ser. No. 18/207,930 and U.S. Provisional Application No. **63/423,911** are assigned to the assignee of the present application.

TECHNICAL FIELD

[0002] The present disclosure relates generally to supply chain planning and specifically to determining inventory availability in a supply chain network.

BACKGROUND

[0003] When computing inventory availability for supply chain products, an inventory availability service may be used, which may be provided via an application programming interface (API). Inventory availability services may encounter sudden spikes and increased processing traffic for a limited set of items due to retailers utilizing various directed marketing methods for a small set of items and the emergence of shopping bots scraping e-commerce websites for discounts. Existing inventory availability services are not designed for these sudden spikes in traffic for a small set of items. Because of the computation demands of existing inventory availability services, it is not uncommon for these inventory availability services to become unresponsive when faced with traffic spikes, resulting in sluggish website response times, or inaccurate inventory availability information, which may result in a retailer promising to sell inventory it does not actually possess, or losing sales due to marking items as out of stock when they are not. Existing inventory availability services are thus not resilient when faced with sudden traffic spikes, resulting in high-latency communications and inaccurate inventory information, both of which are undesirable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] A more complete understanding of the present invention may be derived by referring to the detailed description when considered in connection with the following illustrative figures. In the figures, like reference numbers refer to like elements or acts throughout the figures.

[0005] FIG. 1 illustrates a supply chain network, in accordance with a first embodiment;

[0006] FIG. 2 illustrates the inventory availability system, the archiving system, and the planning and execution system of FIG. 1 in greater detail, in accordance with an embodiment;

[0007] FIG. 3 illustrates a method for routing inventory availability queries, in accordance with an embodiment;

[0008] FIG. 4 illustrates an architecture diagram of the inventory availability system of FIG. 1, in accordance with an embodiment; and

[0009] FIG. 5 illustrates a method for handling partition overload in e-commerce available to promise inquiries, in accordance with an embodiment.

DETAILED DESCRIPTION

[0010] Aspects and applications of the invention presented herein are described below in the

drawings and detailed description of the invention. Unless specifically noted, it is intended that the words and phrases in the specification and the claims be given their plain, ordinary, and accustomed meaning to those of ordinary skill in the applicable arts.

[0011] In the following description, and for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the various aspects of the invention. It will be understood, however, by those skilled in the relevant arts, that the present invention may be practiced without these specific details. In other instances, known structures and devices are shown or discussed more generally in order to avoid obscuring the invention. In many cases, a description of the operation is sufficient to enable one to implement the various forms of the invention, particularly when the operation is to be implemented in software. It should be noted that there are many different and alternative configurations, devices and technologies to which the disclosed inventions may be applied. The full scope of the inventions is not limited to the examples that are described below.

[0012] As described below, embodiments of the following disclosure provide a system and method for routing inventory availability requests to avoid system overload due to a spike in such requests. Embodiments may perform dynamic histogram-based determinations of hot partitions of a distributed cache and accordingly implement dynamic routing of inventory availability traffic. Embodiments may also move items between local caches to increase resource optimization and avoid issues related to high demand items. Embodiments may use different cache expiry periods for different items based on inventory availability levels.

[0013] Embodiments of the following disclosure enable inventory availability systems to be resilient to periods of high request traffic. Use of embodiments may reduce a workload on computing resources during periods of unexpected or undesired spikes in availability requests. Embodiments may enable inventory availability systems to remain in an online state even during periods of extreme traffic spikes. The increased resiliency of embodiments may provide more accurate inventory estimates than existing inventory availability systems due to a higher uptime in such systems.

[0014] FIG. 1 illustrates supply chain network **100**, in accordance with a first embodiment. Supply chain network **100** comprises inventory availability system **110**, archiving system **120**, planning and execution system **130**, one or more supply chain entities **140**, computer **150**, network **160**, and one or more communication links **162-170**. Although a single inventory availability system **110**, a single archiving system **120**, a single planning and execution system **130**, one or more supply chain entities **140**, a single computer **150**, a single network **160**, and one or more communication links **162-170** are shown and described, embodiments contemplate any number of inventory availability systems, archiving systems, planning and execution systems, supply chain entities, computers, networks, or communication links, according to particular needs.

[0015] In one embodiment, inventory availability system **110** comprises one or more API server nodes **112** and distributed cache **114**. Inventory availability system **110** may compute and expose inventory data **266** (FIG. 2) to respond to available to promise (ATP) requests. As described in more detail below, inventory availability system **110** may, in an embodiment, utilize one or more software modules of API server nodes **112** to track traffic data **204** to one or more partitions **220** of distributed cache **114**. Depending on a current status of one or more partitions **220** of distributed cache **114**, including data to a particular item that is the subject of an ATP request, inventory availability system **110** may route the traffic of the ATP request to local cache **216** of one or more API server nodes **112**.

[0016] Archiving system **120** comprises server **122** and database **124**. Although archiving system **120** is shown as comprising a single server **122** and a single database **124**, embodiments contemplate any suitable number of servers or databases internal to or externally coupled with archiving system **120**. Server **122** of archiving system **120** may support one or more processes for receiving and storing data from planning and execution system **130**, one or more supply chain

entities **140**, and/or one or more computers **150** of supply chain network **100**. According to some embodiments, archiving system **120** comprises an archive of data received from planning and execution system **130**, one or more supply chain entities **140**, and/or one or more computers **150** of supply chain network **100**. Archiving system **120** provides archived data to inventory availability system **110** and/or planning and execution system **130**. Server **122** may store the received data in database **124**. Database **124** of archiving system **120** may comprise one or more databases or other data storage arrangements at one or more locations, local to, or remote from, server **122**.

[0017] According to an embodiment, planning and execution system **130** comprises server **132** and database **134**. Supply chain planning and execution is typically performed by several distinct and dissimilar processes, including, for example, assortment planning, demand planning, operations planning, production planning, supply planning, distribution planning, execution, pricing, forecasting, transportation management, warehouse management, inventory management, fulfillment, procurement, and the like. Server **132** of planning and execution system **130** comprises one or more modules, such as, for example, an order capture module, a sourcing module, a scheduling module, and/or a pick-pack-ship module for performing one or more order fulfillment processes. Server **132** stores and retrieves data from database **134** or one or more locations in supply chain network **100**. In addition, planning and execution system **130** operates on one or more computers **150** that are integral to or separate from the hardware and/or software that support archiving system **120** and inventory availability system **110**.

[0018] One or more supply chain entities **140** may include, for example, one or more retailers, distribution centers, manufacturers, suppliers, customers, and/or similar business entities configured to manufacture, order, transport, or sell one or more products. Retailers may comprise any online or brick-and-mortar store that sells one or more products to one or more customers. Manufacturers may be any suitable entity that manufactures at least one product, which may be sold by one or more retailers. Suppliers may be any suitable entity that offers to sell or otherwise provides one or more items (i.e., materials, components, or products) to one or more manufacturers. Although one example of supply chain network **100** is illustrated and described, embodiments contemplate any configuration of supply chain network **100**, without departing from the scope described herein.

[0019] As shown in FIG. **1**, supply chain network **100** comprising inventory availability system **110**, archiving system **120**, planning and execution system **130**, and one or more supply chain entities **140** may operate on one or more computers **150** that are integral to or separate from the hardware and/or software that support inventory availability system **110**, archiving system **120**, planning and execution system **130**, and one or more supply chain entities **140**. One or more computers **150** may include any suitable input device **152**, such as a keypad, mouse, touch screen, microphone, or other device to input information. Output device **154** may convey information associated with the operation of supply chain network **100**, including digital or analog data, visual information, or audio information. One or more computers **150** may include fixed or removable computer-readable storage media, including a non-transitory computer-readable medium, magnetic computer disks, flash drives, CD-ROM, in-memory device, or other suitable media to receive output from and provide input to supply chain network **100**.

[0020] One or more computers **150** may include one or more processors **156** and associated memory to execute instructions and manipulate information according to the operation of supply chain network **100** and any of the methods described herein. In addition, or as an alternative, embodiments contemplate executing the instructions on one or more computers **150** that cause one or more computers **150** to perform functions of the method. An apparatus implementing special purpose logic circuitry, for example, one or more field-programmable gate arrays (FPGA) or application-specific integrated circuits (ASIC), may perform functions of the methods described herein. Further examples may also include articles of manufacture including tangible non-transitory computer-readable media that have computer-readable instructions encoded thereon, and the

instructions may comprise instructions to perform functions of the methods described herein.

[0021] In addition, or as an alternative, supply chain network **100** may comprise a cloud-based computing system having processing and storage devices at one or more locations, local to, or remote from inventory availability system **110**, archiving system **120**, planning and execution system **130**, and one or more supply chain entities **140**. In addition, each of one or more computers **150** may be a workstation, personal computer (PC), network computer, notebook computer, tablet, personal digital assistant (PDA), cell phone, telephone, smartphone, wireless data port, augmented or virtual reality headset, or any other suitable computing device. In an embodiment, one or more users may be associated with inventory availability system **110** and archiving system **120**.

[0022] In one embodiment, inventory availability system **110** may be coupled with network **160** using communication link **162**, which may be any wireline, wireless, or other link suitable to support data communications between inventory availability system **110** and network **160** during operation of supply chain network **100**. Archiving system **120** may be coupled with network **160** using communication link **164**, which may be any wireline, wireless, or other link suitable to support data communications between archiving system **120** and network **160** during operation of supply chain network **100**. Planning and execution system **130** may be coupled with network **160** using communication link **166**, which may be any wireline, wireless, or other link suitable to support data communications between planning and execution system **130** and network **160** during operation of supply chain network **100**. One or more supply chain entities **140** may be coupled with network **160** using communication link **168**, which may be any wireline, wireless, or other link suitable to support data communications between one or more supply chain entities **140** and network **160** during operation of supply chain network **100**. Computer **150** may be coupled with network **160** using communication link **170**, which may be any wireline, wireless, or other link suitable to support data communications between computer **150** and network **160** during operation of supply chain network **100**. Although communication links **162-170** are shown as generally coupling inventory availability system **110**, archiving system **120**, planning and execution system **130**, one or more supply chain entities **140**, and computer **150** to network **160**, any of inventory availability system **110**, archiving system **120**, planning and execution system **130**, one or more supply chain entities **140**, and computer **150** may communicate directly with each other, according to particular needs.

[0023] In another embodiment, network **160** includes the Internet and any appropriate local area networks (LANs), metropolitan area networks (MANs), or wide area networks (WANs) coupling inventory availability system **110**, archiving system **120**, planning and execution system **130**, one or more supply chain entities **140**, and computer **150**. For example, data may be maintained locally to, or externally of, inventory availability system **110**, archiving system **120**, planning and execution system **130**, one or more supply chain entities **140**, and computer **150** and made available to one or more associated users of inventory availability system **110**, archiving system **120**, planning and execution system **130**, one or more supply chain entities **140**, and computer **150** using network **160** or in any other appropriate manner. For example, data may be maintained in a cloud database at one or more locations external to inventory availability system **110**, archiving system **120**, planning and execution system **130**, one or more supply chain entities **140**, and computer **150** and made available to one or more associated users of inventory availability system **110**, archiving system **120**, planning and execution system **130**, one or more supply chain entities **140**, and computer **150** using the cloud or in any other appropriate manner. Those skilled in the art will recognize that the complete structure and operation of network **160** and other components within supply chain network **100** are not depicted or described. Embodiments may be employed in conjunction with known communications networks and other components.

[0024] FIG. 2 illustrates inventory availability system **110**, archiving system **120**, and planning and execution system **130** of FIG. 1 in greater detail, in accordance with an embodiment. Inventory availability system **110** may comprise one or more API server nodes **112** and distributed cache **114**,

as described above. Although inventory availability system **110** is shown as comprising a plurality of API server nodes **112** and a single distributed cache **114**, embodiments contemplate any suitable number of servers or databases internal to or externally coupled with inventory availability system **110**.

[0025] API server node **112** is a node in a distributed computing system, such as a virtual machine (VM), which may be used to perform tasks of inventory availability system **110**. Although in this example multiple API server nodes **112** are shown within inventory availability system **110**, API server nodes **112** may be distributed across multiple physical machines. API server nodes **112** may comprise allocated computing resources, such as, for example, processing time, RAM, database storage, or the like from a physical computing device. According to embodiments, API server nodes **112** may execute independently from other API server nodes **112**, regardless of whether API server nodes **112** are allocated computing resources on the same physical computing device. In embodiments, a load balancer, such as load balancer **404** (FIG. 4), may centrally control API server nodes **112** via a hypervisor layer or other control mechanism to distribute traffic to API server nodes **112** and allocate computing resources to API server nodes **112**.

[0026] API server node **112** of inventory availability system **110** comprises communication module **202**, traffic data **204**, statistics aggregator module **206**, histogram data **208**, hot item module **210**, partition status data **212**, local cache module **214**, and local cache **216**. Although API server node **112** is shown and described as comprising a single communication module **202**, traffic data **204**, a single statistics aggregator module **206**, histogram data **208**, a single hot item module **210**, partition status data **212**, a single local cache module **214**, and a single local cache **216**, embodiments contemplate any suitable number or combination of these located at one or more locations, local to, or remote from inventory availability system **110**, such as on multiple servers or computers **150** at one or more locations in supply chain network **100**.

[0027] Distributed cache **114** of inventory availability system **110** may comprise one or more databases or other data storage arrangements at one or more locations, local to, or remote from, API server nodes **112**. Distributed cache **114** is used by API server nodes **112** to obtain inventory data **266**, which may be computed by one or more of API server nodes **112** or received from planning and execution system **130** and stored in distributed cache **114** to allow wider access to computed inventory data **266**. Distributed cache **114** comprises, for example, one or more partitions **220**, which separate inventory data **266** stored by distributed cache **114**. In general, when API server node **112** attempts to access inventory information about an item of supply chain network **100**, the request is routed to a partition of distributed cache **114** associated with that item. Although distributed cache **114** is shown and described as comprising partitions **220**, embodiments contemplate any suitable number or combination of data, located at one or more locations, local to, or remote from, inventory availability system **110**, according to particular needs.

[0028] Communication module **202** receives inventory information requests, such as, for example, ATP requests from other devices within supply chain network **100** and, once the inventory information has been retrieved, responds to the requests. For example, communication module **202** may receive inventory information requests from an e-commerce web server to determine the inventory availability of items for sale on the e-commerce websites. According to embodiments, communication module **202** retrieves the inventory information from local cache module **214**. Communication module **202** may raise an internal event, which may comprise a number of calls for an item subject to an ATP request.

[0029] Traffic data **204** comprises partition-wise traffic data related to the access rates of partitions **220** of distributed cache **114**. API server node **112** tracks the frequency that each individual partition of partitions **220** is accessed. Sudden increases to the rate of information availability requests to a single partition may indicate that a particular item stored within that partition may be the subject of a sudden increase in demand, which may be the result of a sale, targeting marketing, social media promotions, activity of shopping bots, or other factors.

[0030] Statistics aggregator module **206** uses traffic data **204** to prepare histograms indicating request rates for partitions **220**. According to embodiments, statistics aggregator module **206** comprises a histogram builder which creates histograms over a time interval *I*. For example, statistics aggregator module **206** may run at an interval of/seconds within each API server node **112** and create histograms out of all partition-wise traffic data **204** pooled in that interval. In embodiments, interval *I* may be kept low to minimize the risk of a request rate spike occurring completely within interval *I*. Statistics aggregator module **206** saves the histograms generated as histogram data **208**.

[0031] Hot item module **210** determines whether one or more partitions of partitions **220** is in an overload state, which may also be referred to as the “Sizzling Hot Item” (SHI) state. According to embodiments, hot item module **210** uses histogram data **208** to determine whether a hot item threshold has been reached. In some embodiments, the hot item threshold may be reached when the percentile of hits for a small number of partitions **220** exceeds a threshold (such as, for example, *n* partitions account for *P* percentile, where *n* and *P* are initially determined empirically based on what the cluster is able to process). Hot item module **210** may use any appropriate algorithm, such as percentiles, regression, trends, anomaly detection, and the like to detect traffic to one or more items. In addition, or as an alternative, hot item module **210** may use standard deviation or another statistical value to determine whether the hot item threshold has been reached. For example, when each partition has one transaction per second (TPS) per API node server **112**, with a standard deviation of 25% for the hottest partition of partitions **220**, hot item module **210** may detect a partition overload scenario when the standard deviation of the hottest partition from the non-overloaded mean is 10× or 20×. In other embodiments, the hot item threshold may be reached when the total TPS is greater than a threshold *T* that is set, which may prevent this mechanism from being activated at low volumes. When hot item module **210** determines that the hot item threshold has been reached for a partition of partitions **220**, hot item module **210** marks that partition as “overloaded” or as “SHI.” Hot item module **210** saves the status of partitions **220** (normal load or SHI) as partition status data **212**.

[0032] Local cache module **214** synchronously checks the partition ID of each of the item IDs that API node server **112** is requested to fetch. When one or more partitions **220** is in SHI state as indicated by partition status data **212**, then local cache module **214** switches to fetching data from local cache **216** as opposed to fetching from distributed cache **114**. That is, on detecting a partition as SHI, traffic is automatically adjusted to have dynamic routing to local cache **216**. In embodiments, local cache module **214** also manages local cache **216**, such as by setting an expiry time before local cache **216** is purged and traffic rerouted to distributed cache **114**. When the distribution of calls into API server node **112** that is under an SHI scenario returns to a regular rate, API server node **112** reverts to the regular state and stops inspecting the partition ID of every incoming item.

[0033] According to embodiments, local cache **216** is a temporal cache created by local cache module **214** to redirect traffic from an overloaded partition of partitions **220**. In some cases of extreme request spikes, local cache **216** may become overloaded within the set expiry period of local cache **216**. In such cases, local cache module **214** may generate an additional local cache **216** or redirect traffic of local cache **216** of a different API server node **112**, depending on particular need. When initializing local cache **216**, local cache module **214** may fetch inventory data **266** from distributed cache **114** and store the data of the SHI partition in local cache **216**, allowing the SHI partition to be temporarily accessed directly from API server node **112** rather than from distributed cache **114**.

[0034] As discussed above, archiving system **120** comprises server **122** and database **124**. Although archiving system **120** is shown as comprising a single server **122** and a single database **124**, embodiments contemplate any suitable number of servers or databases internal to or externally coupled with archiving system **120**.

[0035] Server **122** of archiving system **120** comprises data retrieval module **230**. Although server **122** is shown and described as comprising a single data retrieval module **230**, embodiments contemplate any suitable number or combination of data retrieval modules located at one or more locations, local to, or remote from archiving system **120**, such as on multiple servers or computers **150** at one or more locations in supply chain network **100**.

[0036] In one embodiment, data retrieval module **230** of archiving system **120** receives historical supply chain data **240** from planning and execution system **130** and one or more supply chain entities **140** and stores received historical supply chain data **240** in archiving system **120** database **124**. According to one embodiment, data retrieval module **230** of archiving system **120** may prepare historical supply chain data **240** for use as the training data of inventory availability system **110** by checking historical supply chain data **240** for errors and transforming historical supply chain data **240** to normalize, aggregate, and/or rescale historical supply chain data **240** to allow direct comparison of data received from different planning and execution systems **130**, one or more supply chain entities **140**, and/or one or more other locations local to, or remote from, archiving system **120**. According to embodiments, data retrieval module **230** may receive data from one or more sources external to supply chain network **100**, such as, for example, weather data, special events data, social media data, calendar data, and the like and stores the received data as historical supply chain data **240**.

[0037] Database **124** of archiving system **120** may comprise one or more databases or other data storage arrangements at one or more locations, local to, or remote from, server **122**. Database **124** of archiving system **120** comprises, for example, historical supply chain data **240**. Although database **124** of archiving system **120** is shown and described as comprising historical supply chain data **240**, embodiments contemplate any suitable number or combination of data, located at one or more locations, local to, or remote from, archiving system **120**, according to particular needs.

[0038] Historical supply chain data **240** comprises historical data received from inventory availability system **110**, archiving system **120**, supply chain planning and execution system **130**, one or more supply chain entities **140**, and/or computer **150**. Historical supply chain data **240** may comprise, for example, weather data, special events data, social media data, calendar data, and the like. In an embodiment, historical supply chain data **240** may comprise, for example, historic sales patterns, prices, promotions, weather conditions and other factors influencing future demand of the number of one or more items sold in one or more stores over a time period, such as, for example, one or more days, weeks, months, years, including, for example, a day of the week, a day of the month, a day of the year, week of the month, week of the year, month of the year, special events, paydays, and the like.

[0039] As discussed above, planning and execution system **130** comprises server **132** and database **134**. Although planning and execution system **130** is shown as comprising a single server **132** and a single database **134**, embodiments contemplate any suitable number of servers or databases internal to or externally coupled with planning and execution system **130**.

[0040] Server **132** of planning and execution system **130** comprises planning module **250** and prediction module **252**. Although server **132** is shown and described as comprising a single planning module **250** and a single prediction module **252**, embodiments contemplate any suitable number or combination of planning modules and prediction modules located at one or more locations, local to, or remote from planning and execution system **130**, such as on multiple servers or computers **150** at one or more locations in supply chain network **100**.

[0041] Database **134** of planning and execution system **130** may comprise one or more databases or other data storage arrangements at one or more locations, local to, or remote from, server **132**. Database **134** of planning and execution system **130** comprises, for example, transaction data **260**, supply chain data **262**, product data **264**, inventory data **266**, inventory policies **268**, store data **270**, customer data **272**, demand forecasts **274**, supply chain models **276**, and prediction models **278**. Although database **134** of planning and execution system **130** is shown and described as

comprising transaction data **260**, supply chain data **262**, product data **264**, inventory data **266**, inventory policies **268**, store data **270**, customer data **272**, demand forecasts **274**, supply chain models **276**, and prediction models **278**, embodiments contemplate any suitable number or combination of data, located at one or more locations, local to, or remote from, planning and execution system **130**, according to particular needs.

[0042] Planning module **250** of planning and execution system **130** works in connection with prediction module **252** to generate a plan based on one or more predicted retail volumes, classifications, or other predictions. By way of example and not of limitation, planning module **250** may comprise a demand planner that generates demand forecasts **274** for one or more supply chain entities **140**. Planning module **250** may generate demand forecasts **274**, at least in part, from predictions and calculated factor values for one or more causal factors received from prediction module **252**. By way of a further example, planning module **250** may comprise an assortment planner and/or a segmentation planner that generates product assortments that match causal effects calculated for one or more customers or products by prediction module **252**, which may provide for increased customer satisfaction and sales, as well as reducing costs for shipping and stocking products at stores where they are unlikely to sell.

[0043] Prediction module **252** of planning and execution system **130** applies samples of transaction data **260**, supply chain data **262**, product data **264**, inventory data **266**, store data **270**, customer data **272**, demand forecasts **274**, and other data to prediction models **278** to generate predictions and calculated factor values for one or more causal factors. As disclosed above, prediction module **252** of planning and execution system **130** predicts a volume Y (target) from a set of causal factors X along with causal factors strengths that describe the strength of each causal factor variable contributing to the predicted volume. According to some embodiments, prediction module **252** generates predictions at daily intervals. However, embodiments contemplate longer and shorter prediction phases that may be performed, for example, weekly, twice a week, twice a day, hourly, or the like.

[0044] Transaction data **260** of planning and execution system **130** database **134** may comprise recorded sales and returns transactions and related data, including, for example, a transaction identification, time and date stamp, channel identification (such as stores or online touchpoints), product identification, actual cost, selling price, sales volume, customer identification, promotions, and or the like. In addition, transaction data **260** is represented by any suitable combination of values and dimensions, aggregated or un-aggregated, such as, for example, sales per week, sales per week per location, sales per day, sales per day per season, or the like.

[0045] Supply chain data **262** may comprise any data of one or more supply chain entities **140** including, for example, item data, identifiers, metadata (comprising dimensions, hierarchies, levels, members, attributes, cluster information, and member attribute values), fact data (comprising measure values for combinations of members), business constraints, goals and objectives of one or more supply chain entities **140**.

[0046] Product data **264** of database **134** may comprise products identified by, for example, a product identifier (such as a Stock Keeping Unit (SKU), Universal Product Code (UPC), or the like), and one or more attributes and attribute types associated with the product ID. Product data **264** may comprise data about one or more products organized and sortable by, for example, product attributes, attribute values, product identification, sales volume, demand forecast, or any stored category or dimension. Attributes of one or more products may be, for example, any categorical characteristic or quality of a product, and an attribute value may be a specific value or identity for the one or more products according to the categorical characteristic or quality, including, for example, physical parameters (such as, for example, size, weight, dimensions, color, and the like).

[0047] Inventory data **266** of database **134** may comprise any data relating to current or projected inventory quantities or states, order rules, or the like. For example, inventory data **266** may comprise the current level of inventory for each item at one or more stocking points across supply

chain network **100**. In addition, inventory data **266** may comprise order rules that describe one or more rules or limits on setting an inventory policy, including, but not limited to, a minimum order volume, a maximum order volume, a discount, and a step-size order volume, and batch quantity rules. According to some embodiments, planning and execution system **130** accesses and stores inventory data **266** in database **134**, which may be used by planning and execution system **130** to place orders, set inventory levels at one or more stocking points, initiate manufacturing of one or more components, or the like in response to, and based at least in part on, a forecasted demand of planning and execution system **130**.

[0048] Inventory policies **268** of database **134** may comprise any suitable inventory policy describing the reorder point and target quantity, or other inventory policy parameters that set rules for inventory availability system **110** and/or planning and execution system **130** to manage and reorder inventory. Inventory policies **268** may be based on target service level, demand, cost, fill rate, or the like. According to embodiments, inventory policies **268** comprise target service levels that ensure that a service level of one or more supply chain entities **140** is met with a set probability. For example, one or more supply chain entities **140** may set a service level at 95%, meaning one or more supply chain entities **140** sets the desired inventory stock level at a level that meets demand 95% of the time. Although a particular service level target and percentage is described, embodiments contemplate any service target or level, such as, for example, a service level of approximately 99% through 90%, a 75% service level, or any suitable service level, according to particular needs. Other types of service levels associated with inventory quantity or order quantity may comprise, but are not limited to, a maximum expected backlog and a fulfillment level. Once the service level is set, inventory availability system **110** and/or the planning and execution system **130** may determine a replenishment order according to one or more replenishment rules, which, among other things, indicates to one or more supply chain entities **140** to determine or receive inventory to replace the depleted inventory. By way of example only and not by way of limitation, an inventory policy for non-perishable goods with linear holding and shorting costs comprises a min./max. (s,S) inventory policy. Other inventory policies **268** may be used for perishable goods, such as fruit, vegetables, dairy, fresh meat, as well as electronics, fashion, and similar items for which demand drops significantly after a next generation of electronic devices or a new season of fashion is released.

[0049] Store data **270** may comprise data describing the stores of one or more retailers and related store information. Store data **270** may comprise, for example, a store ID, store description, store location details, store location climate, store type, store opening date, lifestyle, store area (expressed in, for example, square feet, square meters, or other suitable measurement), latitude, longitude, and other similar data.

[0050] Customer data **272** may comprise customer identity information, including, for example, customer relationship management data, loyalty programs, and mappings between product purchases and one or more customers so that a customer associated with a transaction may be identified. Customer data **272** may comprise data relating customer purchases to one or more products, geographical regions, store locations, or other types of dimensions.

[0051] Demand forecasts **274** of database **134** may indicate future expected demand based on, for example, any data relating to past sales, past demand, purchase data, promotions, events, or the like of one or more supply chain entities **140**. Demand forecasts **274** may cover a time interval such as, for example, by the minute, hour, daily, weekly, monthly, quarterly, yearly, or any other suitable time interval, including substantially in real time. Demand may be modeled as a negative binomial or Poisson-Gamma distribution. According to other embodiments, the model also takes into account shelf-life of perishable goods (which may range from days (e.g. fresh fish or meat) to weeks (e.g. butter) or even months, before any unsold items have to be written off as waste) as well as influences from promotions, price changes, rebates, coupons, and even cannibalization effects within an assortment range. In addition, customer behavior is not uniform but varies throughout the

week and is influenced by seasonal effects and the local weather, as well as many other contributing factors. Accordingly, even when demand generally follows a Poisson-Gamma model, the exact values of the parameters of the model may be specific to a single product to be sold on a specific day in a specific location or sales channel and may depend on a wide range of frequently changing influencing causal factors. As an example only and not by way of limitation, an exemplary supermarket may stock twenty thousand items at one thousand locations. When each location of this exemplary supermarket is open every day of the year, planning and execution system **130** comprising a demand planner needs to calculate approximately 2×10^4 demand forecasts **274** each day to derive the optimal order volume for the next delivery cycle (e.g. three days).

[0052] Supply chain models **276** of database **134** comprise characteristics of a supply chain setup to deliver the customer expectations of a particular customer business model. These characteristics may comprise differentiating factors, such as, for example, MTO (Make-to-Order), ETO (Engineer-to-Order) or MTS (Make-to-Stock). However, supply chain models **276** may also comprise characteristics that specify the supply chain structure in even more detail, including, for example, specifying the type of collaboration with the customer (e.g. Vendor-Managed Inventory (VMI)), from where products may be sourced, and how products may be allocated, shipped, or paid for, by particular customers. Each of these characteristics may lead to a different supply chain model. Prediction models **278** comprise one or more of the trained models used by planning and execution system **130** for predicting, among other variables, pricing, targeting, or retail volume, such as, for example, a forecasted demand volume for one or more products at one or more stores of one or more retailers based on the prices of the one or more products.

[0053] FIG. **3** illustrates method **300** for routing inventory availability queries, in accordance with an embodiment. Method **300** may be performed by an inventory availability system, such as inventory availability system **110** of FIG. **1**. Method **300** proceeds by one or more activities, which although described in a particular order may be performed in one or more permutations, combinations, orders, or repetitions, according to particular needs.

[0054] At activity **302**, hot item module **210** of inventory availability system **110** requests aggregated statistics for partitions **220** of distributed cache **114**. In an embodiment, the aggregated statistics may comprise one or more histograms for traffic data **204** of one or more partitions **220**. Hot item module **210** may track the number of hits per partition within API server node **112** and may use an event-and-aggregator method to conditionally change the state of a partition to be the SHI state.

[0055] At activity **304**, hot item module **210** determines whether a hot item threshold for one or more partitions **220** has been reached by analyzing the aggregated statistics. In an embodiment, the hot item threshold may be based on the standard deviation of the aggregated statistics, as discussed in further detail below. When the hot item threshold has been reached, at activity **306** local cache module **214** of inventory availability system **110** fetches inventory data **266** for items of partition **220** that have crossed the hot item threshold from local cache **216** of API server node **112**. When the hot item threshold has not been reached, at activity **308** local cache module **214** fetches inventory data **266** from distributed cache **114**.

[0056] FIG. **4** illustrates architecture diagram **400** of inventory availability system **110** of FIG. **1**, in accordance with an embodiment. Architecture diagram **400** shows e-commerce website **402** and inventory availability system **110**. Inventory availability system **110** comprises load balancer **404**, one or more API server nodes **112** (in this case, three), and distributed cache **114**. Load balancer **404** balances a traffic load among API server nodes **112** to prevent any one API server node **112** from becoming overloaded. In an embodiment, API server nodes **112** are compute nodes comprising business logic to compute inventory availability data. API server nodes **112** fetch inventory data **266** from distributed cache **114**.

[0057] According to embodiments, traffic from e-commerce website **402** may be sent to inventory

availability system **110**. For example, e-commerce website **402** may transmit an inventory availability request when an item page is loaded, when a customer adds the item to a shopping cart, or when similar actions are performed. The inventory availability request is processed by inventory availability system **110** by sending the request to one of API server nodes **112** via load balancer **404**. Then, as discussed in more detail below, inventory availability system **110** retrieves inventory data **266** to fulfill the inventory availability request from distributed cache **114** or, when the partition of distributed cache **114** associated with the item is in an overloaded or SHI state, from local cache **216** of API server nodes **112**.

[0058] FIG. 5 illustrates method **500** for handling partition overload in e-commerce ATP inquiries, in accordance with an embodiment. Method **500** may be performed by an inventory availability system, such as inventory availability system **110** of FIG. 1. Method **500** proceeds by one or more activities, which although described in a particular order may be performed in one or more permutations, combinations, orders, or repetitions, according to particular needs.

[0059] At activity **502**, communication module **202** of inventory availability system **110** receives an ATP inquiry (or request) for an item of supply chain network **100**. According to embodiments, communication module **202** may receive the ATP inquiry from an e-commerce website at one or more API server nodes **112** of inventory availability system **110**, as disclosed above.

[0060] At activity **504**, communication module **202** checks whether the item ID associated with the item is included in one or more partitions **220** of distributed cache **114** that are currently marked as SHI. When the partition in which the item ID is included is marked as normal load, local cache module **214** fetches inventory data **266** for the item from distributed cache **114** at activity **506**. At activity **508**, communication module **202** raises an internal event with the item ID, which is the partition key in distributed cache **114**. According to embodiments, the internal event corresponds to the number of calls associated with the item ID. At activity **510**, statistics aggregator module **206** of inventory availability system **110** caches the event raised at activity **508** and generates and/or updates a histogram for the partition to determine whether the partition access rate is increasing or decreasing.

[0061] At activity **512**, hot item module **210** of inventory availability system **110** determines whether the distribution of the histogram associated with the partition is skewed, that is, whether the access rates for the partition has exceeded a hot item threshold. When hot item module **210** determines the access rate for the partition is not above the hot item threshold, at activity **514** hot item module **210** marks the partition as a normal load. When hot item module **210** determines the access rate for the partition is above the hot item threshold, at activity **516** hot item module **210** marks the partition as SHI.

[0062] Returning to activity **504**, when the partition is marked as SHI, at activity **518** local cache module **214** attempts to fetch the inventory availability data for the item from local cache **216** of API server node **112**. At activity **520**, local cache module **214** determines whether an entry for the item is in local cache **216**. For example, when the ATP inquiry is the first inquiry for the item since the partition has been marked as SHI, inventory data **266** for the item would not have been populated to local cache **216** yet. For instance, when there are fifty inquiries per second for a particular item within API server node **112**, the first inquiry goes to distributed cache **114** and the remaining forty-nine retrieves the data from local cache **216**. When the entry for the item is within local cache **216**, local cache module **214** returns the fetched value in response to the ATP inquiry at activity **522**.

[0063] When the entry for the item is not within local cache **216**, at activity **524** local cache module **214** fetches inventory data **266** instead from distributed cache **114**. At activity **526**, local cache module **214** populates inventory data **266** for the item into local cache **216** with an expiry of E seconds. E is computed to be a number such that the number of inquiries going to the SHI partition falls within a partition overload tolerance limit defined as P. That is, E may be dynamically extended to use local cache **216** for a longer period when the load is extremely high rather than a

static interval. The duration of E may be computed such that the number of calls going to a SHI partition never exceeds normal load. In addition, or as an alternative, E may be a range. For example, the range may be from EL to EH, where EH is the value used for items with high volumes of ATP requests, EL is the value used for items which are below safety level, with a value EX between EL and EH that is used for items with intermediate volumes of ATP requests. Local cache module **214** may determine the EX value based, at least in part, on the ATP level. That is, the expiry period for items in SHI partitions with high availability may have higher EX values. Although local cache **216** contains a stale value for a period of E seconds, this safety valve ensures that distributed cache **114** does not get overwhelmed, safeguarding the entire cluster and overall system.

[0064] Inventory availability system **110** may define the cache expiry period E on a per-item basis. Existing inventory data systems do not handle item-level data because supply chain catalogs typically have millions of items, which makes computing item-level data computationally demanding. However, in embodiments inventory availability system **110** defines different cache expiry periods for different items by tracking traffic data for partitions of the items rather than the entire catalog.

[0065] Additionally, after activity **524**, inventory availability system **110** proceeds to activity **508**, raising an internal event indicating access to the partition, and continues with method **500** to compute the histogram at activity **510** and determines whether the access rate has exceeded the hot item thresholds for the partition at activity **512**. Inventory availability system **110** also supports an automatic emergency brake, such as, for example, by applying the brake when P exceeds a threshold on “too many inquiries.” In addition, or as an alternative, inventory availability system **110** may apply the emergency brake functionality on approval from an authorized user. For example, a user may issue a command in response to a notification from inventory availability system **110** that the traffic load is too high for inventory availability system **110** to reroute. When the emergency brake functionality is activated, the item is listed as out-of-stock in both local cache **216** and distributed cache **114**, even when an availability calculation determines otherwise, until the item gets removed from SHI status.

[0066] Inventory availability system **110** may use various threshold values in embodiments as discussed in further detail above, including but not limited to the histogram period I, an anomaly detection threshold, the cache expiry period E, and/or a local cache overwhelm threshold. According to embodiments, inventory availability system **110** may automatically update the threshold values using reinforcement learning. By way of example only and not by way of limitation, when inventory availability system **110** initially uses one hundred milliseconds as the value for E and local cache **216** is recreated shortly after expiry after several iterations of using the same E value, inventory availability system **110** may determine that one hundred milliseconds is too short. In this example, inventory availability system **110** may increase the E value using one or more reinforcement learning models.

[0067] Inventory availability system **110** may utilize a combination of separating the traffic by pre-determined high traffic scenarios into a “read system,” as well as a partition overload defense algorithm, such as method **500** described above. The partition overload defense algorithm uses an API-server-node-level cache to store pre-computed ATP values for extremely high traffic items. The partition overload defense algorithm provides an additional layer of defense to protect distributed cache **114**. In addition, or as an alternative, inventory availability system **110** may use traditional bot attack perimeter defense as well as a read system that caches pre-computed ATP values for a limited pattern of predictable high traffic items.

[0068] Reference in the foregoing specification to “one embodiment”, “an embodiment”, or “some embodiments” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all

referring to the same embodiment.

[0069] While the exemplary embodiments have been shown and described, it will be understood that various changes and modifications to the foregoing embodiments may become apparent to those skilled in the art without departing from the spirit and scope of the present invention.

Claims

1. A computer-implemented method for handling partition overload in e-commerce ATP inquiries, comprising: receiving, by a computer comprising a processor and memory, an available-to-promise inquiry for an item of a supply chain network; in response to an access rate for a partition being marked as being at or over a threshold, performing, by the computer: fetching inventory availability data for the item from a local cache based on determining whether an entry for the item is in the local cache; in response to the determining the entry for the item is not in the local cache, fetching the inventory availability data from a distributed cache; and raising an internal event with an item ID of the item, wherein the item ID comprises a partition key in the distributed cache; in response to the access rate for the partition being marked as not being at or over the threshold, performing by the computer: fetching inventory availability data for the item from the distributed cache; caching, by the computer, the raised internal event and updating, by the computer, a histogram for the partition; determining, by the computer, whether a distribution of the histogram associated with the partition indicates access rates for the partition are above the threshold; in response to the determining that the access rates are above the threshold, marking, by the computer, the partition as super hot inventory; and in response to the determining that the access rates are not above the threshold, marking, by the computer, the partition as a normal load.
2. The computer-implemented method of claim 1, wherein the internal event corresponds to a number of calls associated with the item ID.
3. The computer-implemented method of claim 1, wherein the histogram for the partition determines whether a partition access rate is increasing or decreasing.
4. The computer-implemented method of claim 1, wherein the available-to-promise inquiry is received from an e-commerce website at one or more API server nodes.
5. The computer-implemented method of claim 1, further comprising: populating, by the computer, the inventory availability data for the item into the local cache with an expiry time.
6. The computer-implemented method of claim 5, wherein the expiry time is computed such that a number of inquiries to a super hot inventory partition falls within a partition overload tolerance limit.
7. The computer-implemented method of claim 6, wherein the expiry time is configured to be a range.
8. A system for handling partition overload in e-commerce ATP inquiries, comprising: a computer, comprising a processor and memory, the computer configured to: receive an available-to-promise inquiry for an item of a supply chain network; in response to an access rate for a partition being marked as being at or over a threshold, perform: fetching inventory availability data for the item from a local cache based on determining whether an entry for the item is in the local cache; in response to the determining the entry for the item is not in the local cache, fetching the inventory availability data from a distributed cache; and raising an internal event with an item ID of the item, wherein the item ID comprises a partition key in the distributed cache; in response to the access rate for the partition being marked as not being at or over the threshold, perform: fetching inventory availability data for the item from the distributed cache; cache the raised internal event and update a histogram for the partition; determine whether a distribution of the histogram associated with the partition indicates access rates for the partition are above the threshold; in response to the determining that the access rates are above the threshold, mark the partition as super hot inventory; and in response to the determining that the access rates are not above the

threshold, mark the partition as a normal load.

9. The system of claim 8, wherein the internal event corresponds to a number of calls associated with the item ID.

10. The system of claim 8, wherein the histogram for the partition determines whether a partition access rate is increasing or decreasing.

11. The system of claim 8, wherein the available-to-promise inquiry is received from an e-commerce website at one or more API server nodes.

12. The system of claim 8, wherein the computer is further configured to: populate the inventory availability data for the item into the local cache with an expiry time.

13. The system of claim 12, wherein the expiry time is computed such that a number of inquiries to a super hot inventory partition falls within a partition overload tolerance limit.

14. The system of claim 13, wherein the expiry time is configured to be a range.

15. A non-transitory computer-readable medium embodied with software for handling partition overload in e-commerce ATP inquiries, the software when executed: receive an available-to-promise inquiry for an item of a supply chain network; in response to an access rate for a partition being marked as being at or over a threshold, perform: fetching inventory availability data for the item from a local cache based on determining whether an entry for the item is in the local cache; in response to the determining the entry for the item is not in the local cache, fetching the inventory availability data from a distributed cache; and raising an internal event with an item ID of the item, wherein the item ID comprises a partition key in the distributed cache; in response to the access rate for the partition being marked as not being at or over the threshold, perform: fetching inventory availability data for the item from the distributed cache; cache the raised internal event and update a histogram for the partition; determine whether a distribution of the histogram associated with the partition indicates access rates for the partition are above the threshold; in response to the determining that the access rates are above the threshold, mark the partition as super hot inventory; and in response to the determining that the access rates are not above the threshold, mark the partition as a normal load.

16. The non-transitory computer-readable medium of claim 15, wherein the internal event corresponds to a number of calls associated with the item ID.

17. The non-transitory computer-readable medium of claim 15, wherein the histogram for the partition determines whether a partition access rate is increasing or decreasing.

18. The non-transitory computer-readable medium of claim 15, wherein the available-to-promise inquiry is received from an e-commerce website at one or more API server nodes.

19. The non-transitory computer-readable medium of claim 15, wherein the software when executed is further configured to: populate the inventory availability data for the item into the local cache with an expiry time.

20. The non-transitory computer-readable medium of claim 19, wherein the expiry time is computed such that a number of inquiries to a super hot inventory partition falls within a partition overload tolerance limit.
